

Comp 442 Group Assignment 2

“Phish Detector”

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Revision History

Version	Date	Author(s)	Notes
v1.0	6/12	Haley	Google doc created and outline written for members
v1.1	6/12	Full team	All sections started
v1.2	6/13	Full team	Report finished and reviewed for submission

1. Introduction

1.1 Project Overview

Phishing and spam messages are the most common cybersecurity threats encountered by users on a daily basis. These messages often contain fraudulent content, misleading advertisements, malicious links, or attempts to steal sensitive information. The goal of the Phish Detector project is to develop a machine learning-based system capable of automatically identifying spam and phishing messages and distinguishing them from legitimate communications.

By applying machine learning and natural language processing (NLP) techniques, the project aims to improve message filtering and provide users with an additional layer of protection against potentially harmful content.

1.2 Dataset Overview

The dataset used in this project is the Email Spam Detection Dataset obtained from Kaggle. The dataset contains 5,572 labeled messages and is intended for supervised machine learning classification tasks.

The dataset consists of two attributes:

- **Category:** Indicates whether a message is classified as spam or ham (legitimate).
- **Message:** Contains the text content of the message.

The dataset combines structured tabular data with unstructured text data. The category attribute serves as the target variable, while the message attribute provides the textual information used for feature extraction and model training. This dataset is appropriate for spam detection because it contains labeled examples that can be used to train and evaluate machine learning models.

One limitation of the dataset is that some messages contain region-specific language and terminology, which may reduce the model's ability to generalize to all communication environments. However, the dataset remains suitable for supervised spam detection because it contains clearly labeled examples of both spam and legitimate messages.

2. Dataset Exploration

2.1 Dataset Size and Structure

The Email Spam Detection Dataset contains 5,572 records and 2 attributes. The dataset is stored in CSV format and is organized into rows and columns. Each row represents a single message, while each column contains information describing that message.

The dataset consists of structured tabular data at the table level and unstructured text data within the Message attribute. This combination makes the dataset suitable for machine learning and natural language processing tasks.

2.2 Dataset Attributes

The dataset contains the following attributes:

Category

- Indicates whether a message is classified as spam or ham (legitimate).
- Serves as the target variable for supervised machine learning.

Message

- Contains the text content of the message.
- Serves as the primary input feature for classification.

Additional features such as message length, word count, URL count, and TF-IDF values may be created during preprocessing and feature engineering.

2.3 Sample Records

Each record in the dataset consists of messages and its corresponding classification label. Sample records show that spam messages often contain promotional language, urgent requests, prize offers, or suspicious links, while ham messages typically contain normal personal or business communication.

Reviewing sample records helps verify label consistency and provides insight into the patterns that may be useful for classification.

2.4 Missing Values Analysis

A missing value analysis was performed to determine whether the dataset contained incomplete records. The results showed that no missing values were present in either the Category or Message attributes.

- Category: 0 missing values
- Message: 0 missing values

Since no missing values were detected, no imputation or missing value treatment is required during preprocessing.

2.5 Duplicate Records Analysis

Duplicate record analysis identified 415 duplicate messages within the dataset. Duplicate records may introduce bias into machine learning models because repeated messages can cause certain patterns to be overrepresented during training. To improve data quality and reduce potential bias, duplicate records should be reviewed and removed during the preprocessing phase before model development.

3. Class Distribution Analysis

3.1 Class Counts

The dataset contains two classes: ham and spam messages.

The class distribution is as follows:

- Ham: 4,825 messages
- Spam: 747 messages

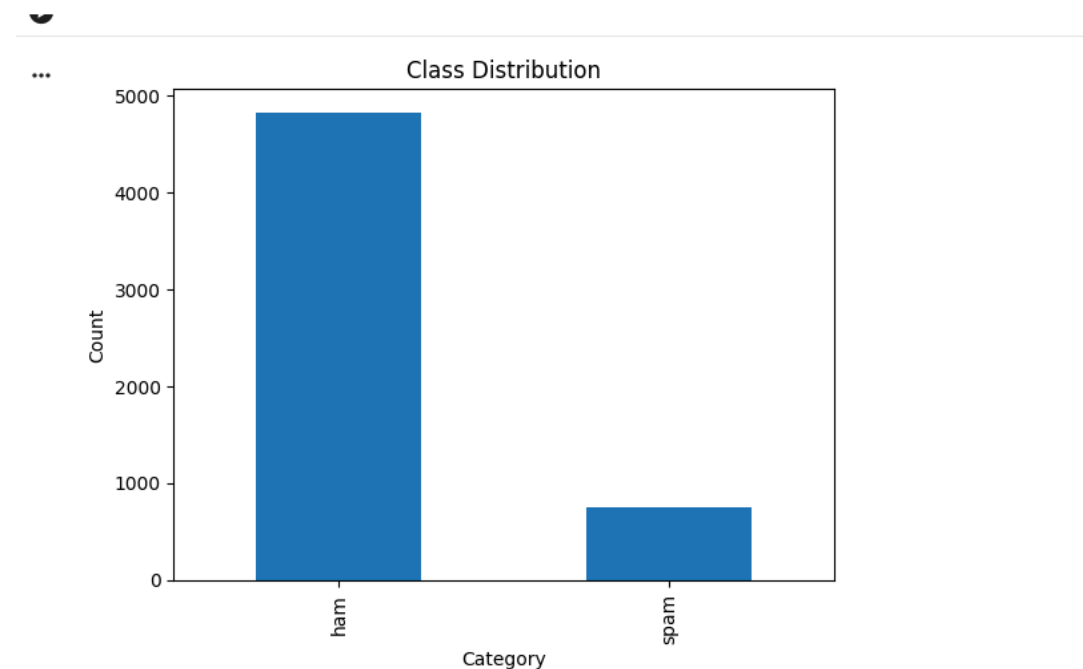
These counts show that legitimate messages make up the majority of the dataset, while spam messages represent a smaller portion of the available data.

3.2 Class Percentages

To better understand the distribution of the classes, percentages were calculated for each category.

- Ham: 86.6%
- Spam: 13.4%

The class distribution chart confirms that ham messages significantly outnumber spam messages within the dataset.



3.3 Class Imbalance Discussion

The dataset exhibits class imbalance because the majority class (ham) contains substantially more records than the minority class (spam). Specifically, approximately 86.6% of the dataset consists of ham messages, while only 13.4% consists of spam messages.

Class imbalance is an important consideration in machine learning because classification models may become biased toward predicting the majority class. As a result, the model may achieve high overall accuracy while failing to correctly identify spam messages.

To address this issue, model performance will be evaluated using multiple metrics such as precision, recall, and F1-score rather than relying solely on accuracy. Additional preprocessing techniques may also be considered if class imbalance negatively affects model performance.

4. Statistical Analysis of Message Length

4.1 Central Tendency

The mean message length was 80.37 characters, indicating the average size of messages in the dataset

The median length was 61 characters, meaning half of all messages contain fewer than 61 characters and half contain more.

The mode of the message length distribution was 22 characters, indicating the most frequently occurring message length.

The midrange of the dataset is $\text{min} + \text{max} / 2 = 2 + 910 / 2 = 456$ characters

4.2 Data Dispersion

The range of message lengths is $910 - 2 = 908$ indicating a large spread between the shortest and longest messages.

The variance of the message length distribution is 3591.24.

The standard deviation is 59.63, indicating substantial variability in message lengths.

Quartiles: $Q1 = 35.75$, $Q2 = 61$, $Q3 = 122$

Interquartile Range (IQR): $\text{IQR} = 122 - 35.75 = 86.25$

4.3 Five-Number Summary

Minimum: 2

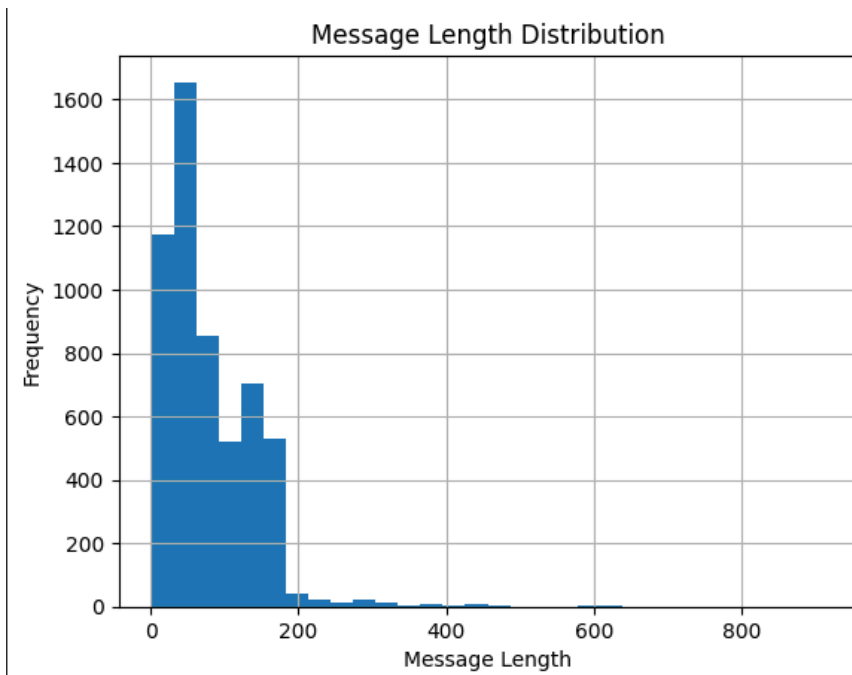
Q1: 35.85

Median: 61

Q3: 122

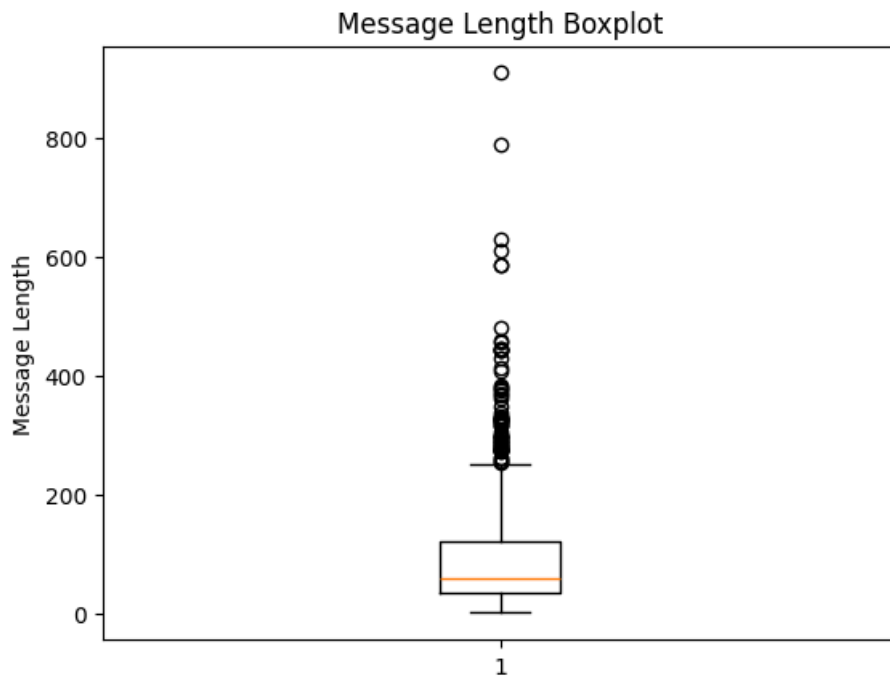
Maximum: 910

4.4 Histogram Analysis



The histogram shows that most messages are relatively short, with the majority of observations concentrated below 200 characters. The distribution contains a small number of unusually long messages, creating a long tail on the right side of the graph.

4.5 Boxplot Analysis

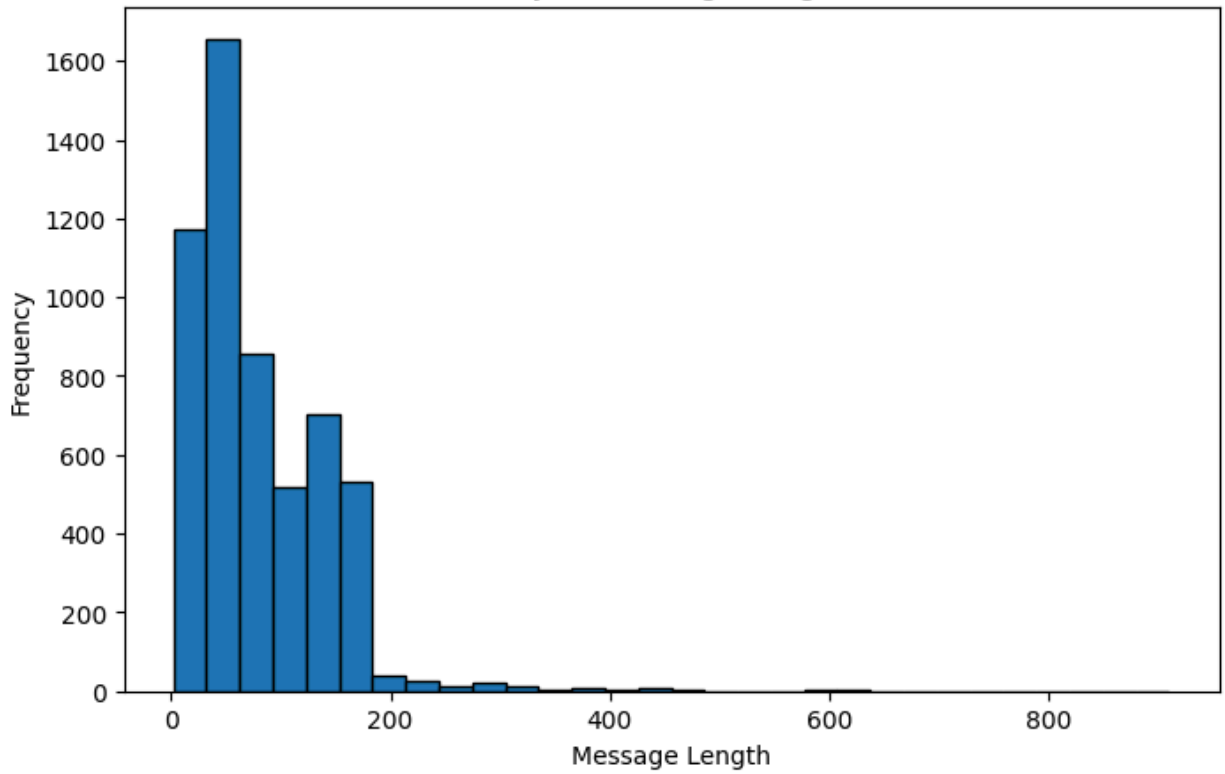


The boxplot shows several outliers above the upper whisker. These outliers correspond to unusually long messages and contribute to the overall spread of the data.

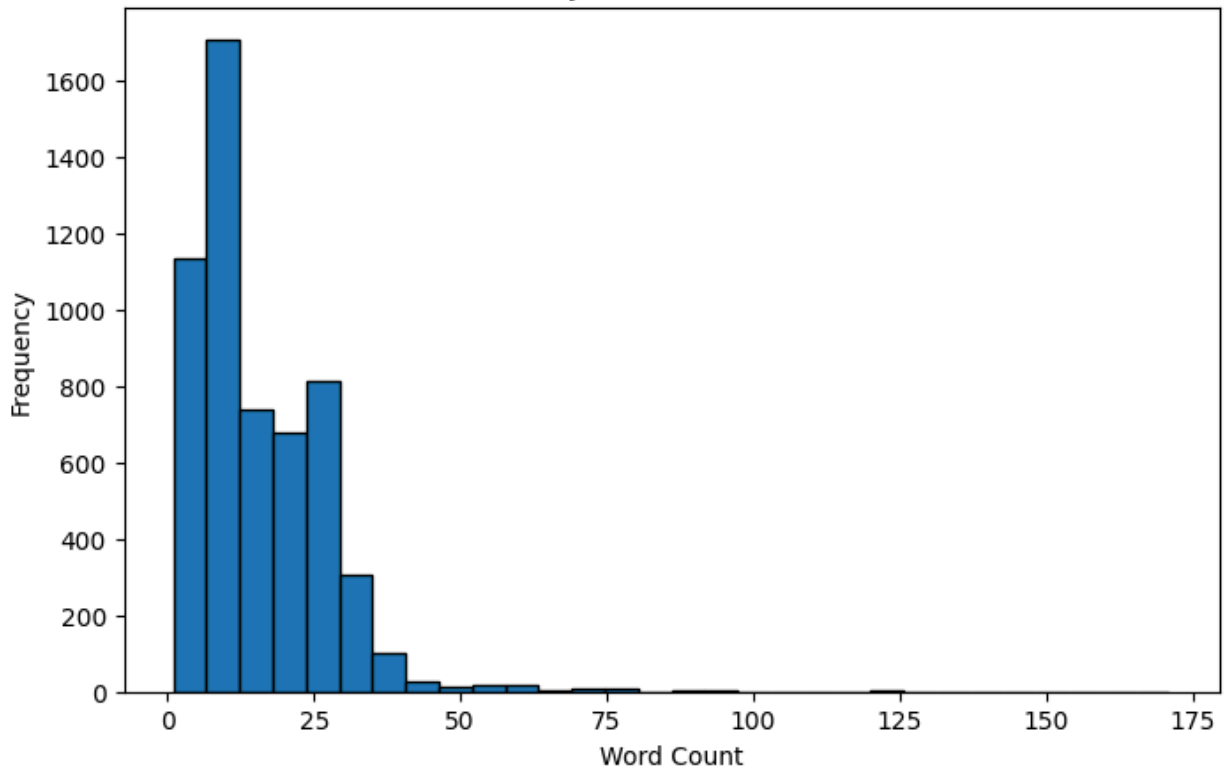
4.6 Skewness Analysis

The skewness value of 2.51 indicates that the distribution is strongly positively skewed. Most messages are relatively short, while a smaller number of messages contain substantially larger amounts of text.

Skewness Analysis: Message Length Distribution



Skewness Analysis: Word Count Distribution



5. Statistical Analysis of Word Count

5.1 Central Tendency

Mean: 15.58

Median: 12

Mode: 6

Midrange: 86

5.2 Data Dispersion

Range: 170

Variance: 130.11

Standard Deviation: 11.41

Quartiles

Q1: 7

Median: 12

Q3: 23

Interquartile Range (IQR): 16

5.3 Five-Number Summary

Min: 1

Q1: 7

Median: 12

Q3: 23

Max: 171

5.4 Skewness Analysis

Skewness: 2.69

6. Correlation Analysis

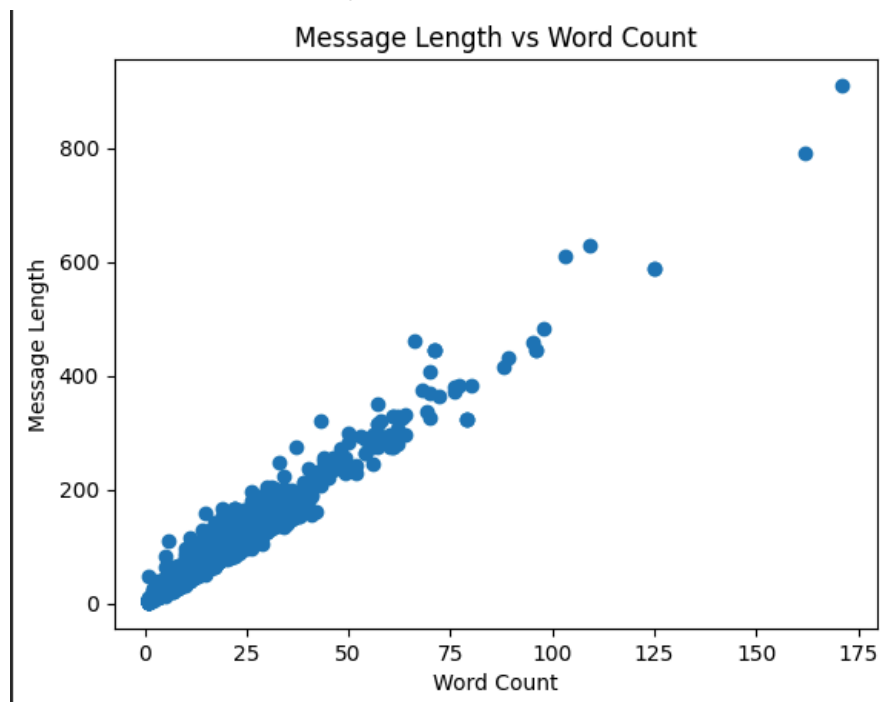
6.1 Message Length and Word Count Correlation

A Pearson correlation analysis was conducted to determine the relationship between message length and word count.

6.2 Correlation Interpretation

The correlation coefficient between message length and word count was 0.974, indicating a very strong positive relationship. As the number of words in a message increases, the overall message length also increases.

6.3 Scatter Plot Analysis



The scatter plot demonstrates a clear positive linear relationship between message length and word count. Messages containing more words tend to have greater overall strength, confirming the strong correlation observed in the statistical analysis.

7. Findings and Observations

7.1 Key Findings

The dataset contains 5,572 total messages with two main columns, category and message. The category column identifies whether the message is ham or spam, while the message column contains the actual text content. The dataset has 4,825 ham messages and 747 spam messages. This means approximately 87% of the dataset is ham and 13% is spam. This shows that our current dataset is imbalanced. There were no messages with missing values in both columns but did contain 415 duplicate records.

Message length and word count were also analyzed. The average message length was 80.37 characters, and average word count was 15.58 words. Both message length and word count were strongly right-skewed, meaning that most messages were short while a few were longer. The correlation between message length and word count was about 0.974, showing a positive relationship.

7.2 Challenges Identified

One challenge identified in the data collection is class imbalance. Since ham messages made up most of the data, a machine learning model may become biased towards predicting ham. This could cause the model to miss spam messages, which would be an issue for an accurate model.

Another challenge is the duplicate records in the data. Duplicate messages can affect the model training because the model may memorize repeated examples instead of learning to generalize patterns. The dataset also contains outliers. Most of the messages were short but a few messages are much longer than the rest. These long messages may affect statistical analysis and may need to be inspected during preprocessing. Text data itself was also challenging due to the lack of structure. Messages may contain punctuation, capitalization differences, abbreviations, numbers, links, and unusual formatting.

Another challenge is that the dataset contains some region-specific language and communication patterns. While these messages still provide useful examples of spam and legitimate content, the vocabulary and phrasing may not fully represent all email communication environments. As a result, this may slightly limit the model's ability to generalize to messages from different regions or user populations.

7.3 Implications for Machine Learning

The findings show that preprocessing will be an important part of this project. Before training the model, the dataset should be cleaned by removing duplicates, standardizing text, and converting message content into numerical features.

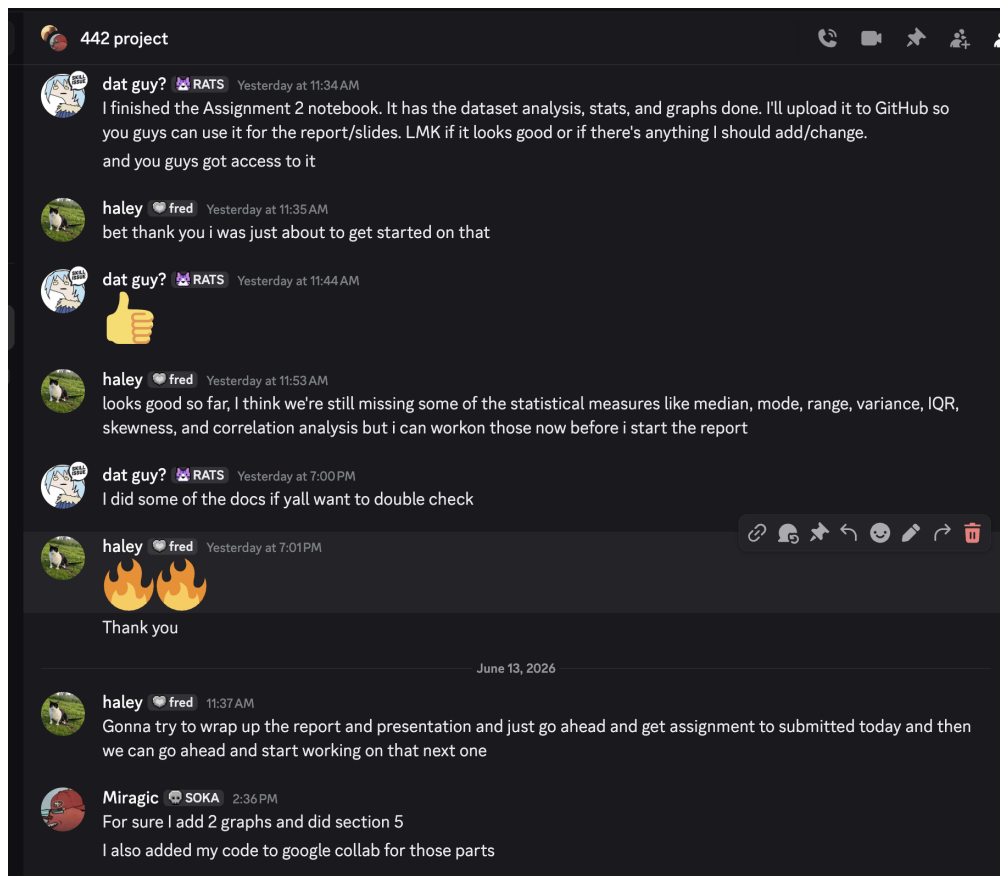
Due to the imbalance of the dataset, the model should not be evaluated solely on accuracy. Precision, recall, F1-score, and a confusion matrix should be used. Recall is important because the model should correctly identify as many spam messages as possible. The strong correlation between messages and words shows that the two features are closely related.

Overall, the data exploration suggests that this dataset is suitable for a supervised machine learning classification task. The next steps are text preprocessing, feature extraction, model training, and evaluation using classification metrics.

8. Conclusion

This assignment explored the email spam detection dataset using statistical analysis techniques to better understand its structure, characteristics, and suitability for machine learning applications. The analysis identified class imbalance, strong positive skewness in both message length and word count, and a strong positive correlation between these two features. The dataset was found to be suitable for supervised machine learning and text classification tasks. These findings provide us valuable insight into the data and will help guide future preprocessing, feature engineering, and model development for the Phish Detector project.

Appendices



Proof of collaboration. We did not meet on call for this assignment.